Antonio Balanzategui

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Experience

Project Engineer — Control Systems Software Jan 2026 – Present
Engineered Services, Inc. (contracted to VCU Health) *Richmond, VA*

- Initial February statistical analysis correctly predicted a mechanical change-point from BAS sensor data alone (CUSUM, Mann-Kendall); validated three months later when the EF1 exhaust fan failed catastrophically on March 19, 2026. VCU Health Pharmacy regulatory compliance leadership formally requested quarterly investigations going forward.
- Write and revise production sequence-of-operations logic on the hospital's BAS (EcoStruxure Building Operation, Andover Continuum) for RTUs, exhaust fans, pumps, pharmacy cleanrooms, and air-compressor failover.
- Programmed a new burn-ICU expansion's rooftop unit and eight VAV boxes from scratch through startup and test-and-balance: an AS-P state-machine program (lockouts, changeover, freeze/smoke shutdown, dual-fan proof, static-pressure reset, min-OA flow) and native MP-V VAV control with a safety guard. Proved every branch in 14 drills on the live controller; fixed a PID anti-windup defect and wrote a custom XPI block that sets each PID's mode from unit state and control mode so idle loops cannot wind up.
- Diagnose and retune unstable control loops with frequency-domain analysis: modeled a chilled-water pump loop as first-order-plus-dead-time, found the $-180°$ phase crossover, and showed the installed gain left a gain margin below 1 — a one-line gain change stopped pumps hunting 25–100% and held ΔP at setpoint in production. In separate self-directed analysis, proved by reachability argument and an MPC benchmark that no controller could hold a fan-coil unit's commanded state, then designed a damped positional PI replacement (ζ 0.13→0.55, ~13× less valve travel in simulation). Both written up as public case studies.
- Built and maintain an automated Python pipeline over 10+ months of BAS trend-log data across 12 sensor channels — change-point detection, multivariate SPC, Hawkes processes, Kalman smoothing, Weibull survival — generating the full quarterly investigation end-to-end and replacing manual analysis.
- Diagnosed a BAS historian stale-data cascade (Hawkes branching ratio $n^* = 0.973$, near-critical) across all 12 channels, tracing it to an ExTL buffer / Maximum Transfer Interval mismatch; the config fix cut worst-case staleness 24 hours → 1.
- Built two internal tools on the MaintainX and Freshdesk REST APIs: one that migrates service tickets out of MaintainX into Freshdesk, replacing manual re-keying of hundreds of tickets; and one that auto-fills service tickets from Freshdesk time-log entries and routes them to sales. Both in use across the campus.

Undergraduate Teaching Assistant, Introduction to GUI Programming Aug 2024 – May 2025
Virginia Tech, Department of Computer Science *Blacksburg, VA*

- Taught 100+ students object-oriented design, event-driven programming, GUI architecture, and the mathematics of 2D graphics.

Projects

EvolutionSim — Real-Time GPU Thunderstorm & Lightning Engine 2025 – Present
Personal research project (private repo); 40-page methods & verification paper and demo available on request

- Built a real-time GPU storm engine (Rust + wgpu + WGSL, ~100 storm compute kernels) simulating deep moist convection coupled to explicit electrification on grids up to 5.3M cells: conservative flux-form (MPDATA) advection, virtual-potential-temperature buoyancy, geometric-multigrid pressure *and* electric-field solvers, two-moment bulk microphysics, and non-inductive graupel–ice charge separation.
- Made the GPU hybrid the production engine for the paper's storm results, with the double-precision CPU implementation kept as a kernel-parity oracle (per-kernel gates at ~10^{-6} relative RMS on the gated build, i.e. single-precision tolerance); profiled the flash handler — the host round-trip I meant to optimize was 1.6% of per-flash time, the multigrid re-solve 67% — then parallelized that solve's red-black smoother across 16 cores for a 1.9× *bit-identical* speedup, rejecting a faster variant that perturbed the result. 5–8× the single-threaded reference at matched resolution; four real-sounding storms complete in 6–18 min each in one unattended overnight queue.
- Made lightning emergent rather than scripted: field-directed stepped leaders grow along the resolved electric field, and each flash is typed cloud-to-ground / intracloud / cloud-air by where its channel terminates — the charge tripole, lightning jump, and multicell self-organization all arise from the physics; flash rates land within 1.38–1.86× of Price–Rind.
- Checked against observation: total-water conservation to machine precision on the f64 reference (10^{-9} relative on the GPU); 14 real radiosonde soundings (CAPE 0–4822 J/kg) reproducing the continental-vs-maritime updraft contrast; and two severe-storm soundings — matching the Norman, OK tornadic storm's NEXRAD echo top to ~4% (15.3 vs ~15.9 km), and a Dodge City storm whose depth, intensity and electrification are consistent with the large hail reported in the same air mass; an automated 9-check capability scorecard scores every run from saved output alone, and a six-seed ensemble separates the exactly-deterministic fluid/microphysics solve from the one stochastic component (mean flash rate robust to ~4%). Part of a 371-mode platform with an ~10,000-test pytest suite.

Education

Bachelor of Science, Computer Science May 2025
Virginia Tech, Blacksburg, VA

- GPA: 3.3 | Pollard Scholarship | 2nd Place, VTURCS Undergraduate Research Symposium — de novo protein-structure prediction from sequence via simulated annealing in AMBER (Scientific Computing).

Skills

Languages: Python, Rust, C++, Java, JavaScript, TypeScript, SQL, C, R, MATLAB; Spanish (proficient).
GPU & Graphics: wgpu, WGSL, compute shaders, GPU-accelerated PDE solvers, CPU↔GPU parity validation, volumetric rendering.
Scientific Computing & Physics: Navier-Stokes (Boussinesq, Chorin projection), conservative flux-form advection (MPDATA), geometric multigrid PDE solvers, stencil methods on structured grids, two-moment bulk microphysics, sub-grid turbulence closures (Deardorff), CRM verification (Kessler, Weisman–Klemp), molecular dynamics (AMBER).
ML, Statistics & Data: NumPy, SciPy, pandas, scikit-learn, PyTorch, LightGBM, matplotlib, data pipelines; CUSUM, PELT, Mann-Kendall, MEWMA Hotelling T-squared, Hawkes processes, Kalman/RTS smoothing, Weibull survival, PCA.
Software & Systems: pytest, parity harnesses & golden-state regression, profiling, REST APIs, Docker, Linux, Git, PostgreSQL, LaTeX.
Control & BAS: PID/PI tuning, frequency-domain stability (gain & phase margin), dead-time compensation, anti-windup, MPC; EcoStruxure, Andover Continuum, BACnet/IP & MSTP.